8. Python Program to Implement DFS (Depth-First Search)

Aim: To implement Depth-First Search (DFS) in Python to traverse a graph by exploring one path as deeply as possible before backtracking.

Procedure:

1. Define a graph using an adjacency list.
2. Select the starting node.
3. Create a set to store visited nodes.
4. Visit the current node and mark it as visited.
5. Recursively visit each unvisited neighbouring node.
6. Continue until all reachable nodes are visited and display the traversal order.

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

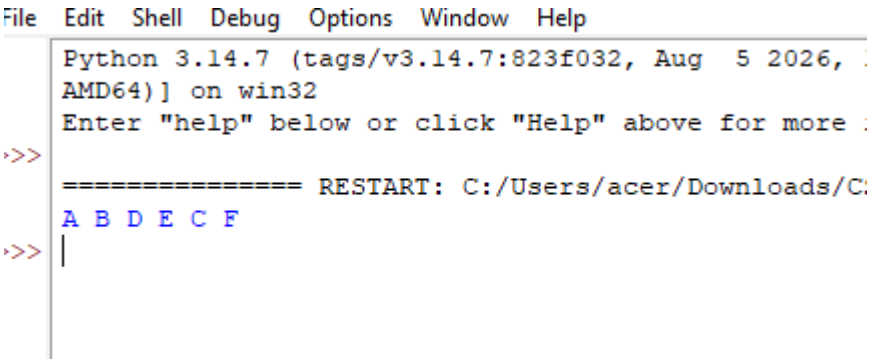
visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbour in graph[node]:
            dfs(neighbour)

dfs('A')
```

Output:



Result: Thus, Depth-First Search was implemented successfully using Python, and the graph was traversed using depth-first exploration.